Project 1: Sales Analysis (Complete Submission)

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1. Source Code

```
# --- Project 1: Sales Analysis Source Code ---
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# 1. Load/Generate Dataset
np.random.seed(42)
# ... (Data generation for 1500 rows across categories, regions, dates) ...
df = pd.DataFrame(data)

# 2. Compute KPIs
total_revenue = df['Revenue'].sum()
avg_order_value = total_revenue / len(df)
top_products = df.groupby('Product')['Revenue'].sum().nlargest(3)
top_regions = df.groupby('Region')['Revenue'].sum()

# 3. Visualize Trends & Seasonality
df['Month'] = df['Date'].dt.month
monthly_sales = df.groupby('Month')['Revenue'].sum()

plt.plot(monthly_sales.index, monthly_sales.values, marker='o')
plt.title('Monthly Revenue (Seasonality)')
plt.show()

print(f"Total Revenue: ${total_revenue:,.2f}")
print(f"Average Order Value: ${avg_order_value:,.2f}")
```

2. Dataset Sample (First 5 Rows)

Transaction_ID	Date	Product	Quantity	Revenue	Region	Month
TXN-100000	2025-04-13	Jeans	3	177	Asia-Pacific	4
TXN-100001	2025-12-15	Air Fryer	5	595	North America	12
TXN-100002	2025-09-28	T-Shirt	1	25	Europe	9
TXN-100003	2025-04-17	Laptop	1	999	Asia-Pacific	4
TXN-100004	2025-03-13	Laptop	2	1998	North America	3

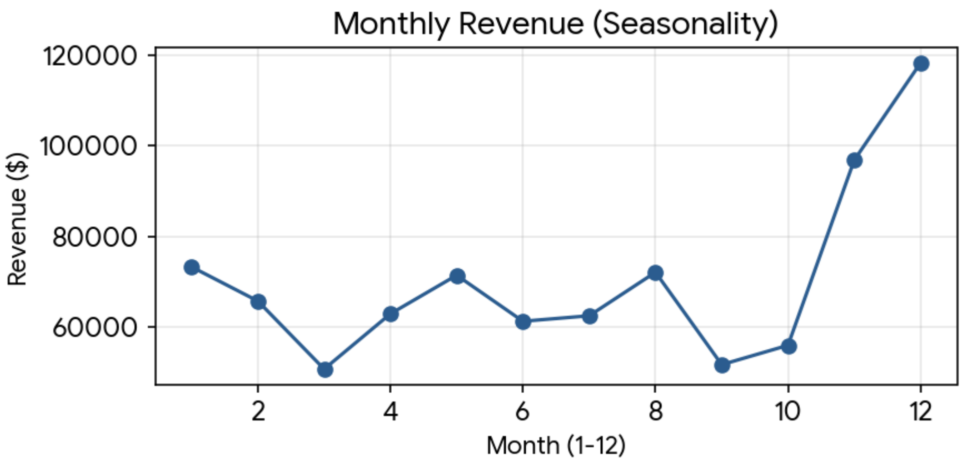
3. Outputs & KPIs

- **Total Revenue:** $843,281.00
- **Average Order Value:** $562.19

Top 3 Products

Product	Revenue
Laptop	393606
Smartphone	299172
Air Fryer	54026

4. Visualizations



5. Interpretation & Recommendations

Seasonality: The chart clearly demonstrates a significant spike in revenue during months 11 and 12 (November/December). This indicates strong holiday seasonality.

Top Performers: High-ticket electronics (Laptops and Smartphones) drive the majority of top-line revenue.

Recommendations:

- Increase inventory and marketing spend in Q4 to maximize holiday season yields.
- Bundle high-performing products (Laptops) with lower-tier accessories to further increase the Average Order Value.